

## Section 1: Landmark Trials in SGLT2 Inhibition and Heart Failure

EMPA-REG OUTCOME and DAPA-HF Clinical Summary: Large randomized double-blind placebo-controlled trials demonstrated that SGLT2 inhibitors provide profound cardiovascular risk reduction.

In the DAPA-HF trial (4,744 patients with HFrEF), Dapagliflozin reduced the primary composite outcome of cardiovascular death or worsening heart failure by 26% (HR 0.74; 95% CI 0.65-0.85;  $P < 0.001$ ) compared with placebo. Benefits were statistically identical in patients with and without Type 2 diabetes.

## Section 2: Comparative Efficacy of Incretin-Based Therapies

Head-to-head clinical evaluation of GLP-1 Receptor Agonists (Semaglutide, Tirzepatide) versus SGLT2 inhibitors indicates greater weight loss and HbA1c reduction with GLP-1 RAs (mean HbA1c reduction 1.5% to 2.2% vs 0.8% to 1.1% with SGLT2i).

Conversely, SGLT2 inhibitors demonstrate superior rapid reductions in heart failure hospitalizations and acute decongestion benefits in cardiorenal syndromes.